

Root phone (POCO X5 5G)

Privacy Analysis of Augmented Reality Glasses

1 Unlocking the Bootloader

Note that Xiaomi provides official USB drivers for Windows, which are required for both fastboot and Mi Flash Tool to properly detect the phone. Other systems do not have native drivers for Xiaomi devices, and needs to be configured but can still have compatibility issues, therefore a Windows computer is what was used (and is preferred) for the unlocking the bootloader.

Steps to unlock bootloader

On the phone:

- Make sure to have Xiaomi account that is linked to the phone number of the sim card and that the number has mobile data. The steps to creating this account can be found in the section below.
- Become a developer by going to 'About phone' and tapping 'MIUI version' 7 times until the following message appears: You are now a developer.
- Go to 'Additional settings', 'Developer options',
- Enable USB debugging and OEM unlocking.
- Turn off wifi and enable cellular data from the sim card.
- Go back to 'additional settings', 'developer options' and go to 'Mi unlock status'.
- Agree to the permissions and press 'Add account and device'. It should a successful message that you were able to add the device at the bottom of the screen.

Now go onto the laptop:

- Install Mi Unlock Tools on the laptop, need windows (unofficial Mi Unlock tool).
- Extract the file you downloaded.
- Double tap MiUsbDriver from the extracted folder to extract.
- Double tap MiUsbDriver to launch application.
- You should see that the opened window says that phone is not connected.

Now back onto the phone:

- Power off the phone.
- With the Xiaomi Poco X5 powered off, press and hold the 'Volume Down' + 'Power' buttons until you see the Xiaomi Poco X5 Fastboot mode screen.
- Connect the Xiaomi Poco X5 phone to your PC / laptop with an original USB cable.

Now onto the laptop:

- In your opened window, it should now start to install drivers.
- After the installation is completed, go back to the folder for the Mi Unlock Tools.
- Double tap miflash_unlock to extract.

- Double tap `miflash_unlock` to launch application.
- Log in to mi unlock (better to use phone number for log in info so they don't ask for verification with phone number, hence won't need to reboot your phone then power it off and put into fastboot mode again)
- App should show: Phone connected. Then press 'Unlock'.
- You should see the window saying 'Unlocked successfully'. Then press 'Reboot phone'.
- After setting up the phone, go to 'Mi Unlock Status' in settings (remember you will need to make yourself a developer again), it should show that the device is unlocked.

2 Rooting the phone

Something to be cautious is that although TWRP is a very common approach for rooting a phone, it did not work for me. Usually, the rooting process of a device with an unlocked bootloader would involve installing a custom recovery such as TWRP and then flashing Magisk directly as a ZIP file. However, the phone I am using uses dynamic partitions, and TWRP had trouble working with these partitions, and couldn't mount (or access) some parts of the system like `'/data'` or `'/storage'`. This caused errors like:

```
"Unable to mount '/data'"
"Repacking image failed."
```

Therefore, I chose another method by patching and flashing the `'boot.img'` with Magisk directly, therefore bypassing the need for TWRP.

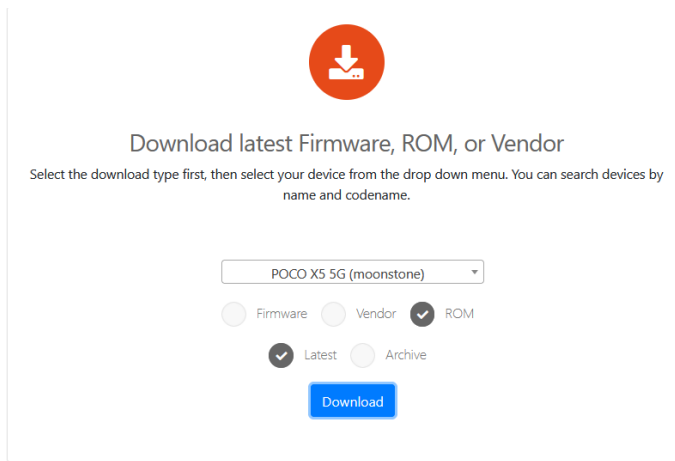
Note that Xiaomi provides official USB drivers for Windows, which are required for both **fastboot** and **Mi Flash Tool** to properly detect the phone. Other systems do not have native drivers for Xiaomi devices, and needs to be configured but can still have compatibility issues. Therefore a Windows computer is what was used (and is preferred) for the rooting the phone.

I also used an SD card, so if you don't have an SD card holder for the phone, this method is not viable. If there is no SD card holder on the computer, you can use another one to transfer files to the SD card.

Steps to root phone

On the computer that has an SD card holder:

- Plug the SD card into your computer.
- Install the latest stable Magisk APK from the link: <https://github.com/topjohnwu/Magisk/releases>
- Transfer the Magisk apk onto the SD card. You can do this by directly dragging and dropping the file onto the mounted SD card in file explorer.
- Download the fastboot ROM (it is also in case of any issues with the phone where any partition or files get corrupted, so the phone becomes dead and you are unable to reboot the phone). This allows you to restore the phone. It can be found with the following link: <https://xmfirmwareupdater.com/>.
- In the window choose your device model and the rest like so



- Press '**Download**' on the window
- On the new page, find a '**device**' (it should be the device you selected but you need to choose a region) that is closest to your location, mine was '**EEA**'. Make sure that the '**branch**' is '**stable**' and the '**type**' is '**Fastboot**'. Then press download.
- Scroll down in the new window and choose one of the displayed '**Full ROM Mirror #**' to download. If one takes more than half an hour, switch to another one as they are simply different services for the same download.
- After download, extract the .tgz file, then extract the .tar file inside it. For Windows, you may need to install '**7-zip**' to extract the files.
- In the extracted folder, go into '**images**' and transfer '**boot.img**' onto the SD card.

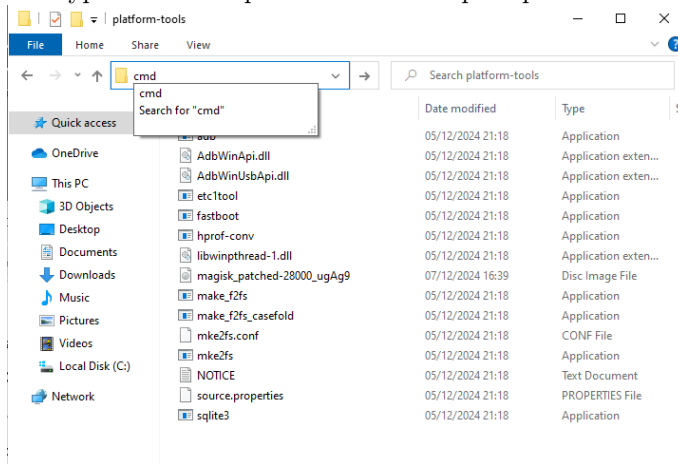
On the phone:

- Make sure that you are a developer in settings. To do this, tap the build number 7 times in '**About phone**'.
- Then make sure to have USB debugging enabled in '**Additional settings**', then '**Developer options**'.
- Plug the SD card into the phone.
- Go to '**File Manager**' and go to where the '**Magisk APK**' is located (it should be in storage/sdcard).
- Tap on the APK to install it, and if prompted enable '**Install Unknown Apps**' in the phone settings. This should start the download of the app on your phone.
- Open the app, and under the word '**Magisk**', you should see something along the lines: '**Magisk is not installed**'.
- Tap '**Install**' to the right side of '**Magisk**'. Then '**Select and patch a file**'.
- Go to the location of the '**boot.img**' on the phone (it should be in storage/sdcard), where Magisk will then process and create a patched version of the file. This file will be saved as '**magisk_patched.img**' in the '**Downloads**' folder of the phone.
- Connect the phone to the windows computer and make sure that it is in 'File Transfer' mode.

On the Windows computer:

- Install '**android platform tools**' (adb and fastboot). Note that if you unlocked the bootloader using the steps from the section '**Unlocking the Bootloader**', this should already be installed and you can directly go to the next step. If not, install it from the link: <https://www.getdroidtips.com/download-adb/#For-Windows>. Then extract the downloaded zip file.
- Go to the extracted '**platform tools**' folder.

- On the top left corner of '**File Explorer**' where the path to the folder is shown, tap in there and type '**cmd**' to open the command prompt in in that directory. Like so



- Check if the connection to the phone is successful by writing the following in the command prompt:

```
adb devices
```

You should see your phone. If not, recheck the USB connectivity.

- Copy the patched magisk file from your phone to your PC using the command:

```
adb pull /sdcard/Download/magisk_patched.img
```

Note that the file will be saved in the same folder as the current directory in the command prompt.

- The '**magisk_patched.img**' and your '**fastboot**' executable from the platform-tools directory should be in the same directory.

Now on the phone:

- Power off the phone.
- Go into fastboot mode by pressing and holding the '**Power**' + '**Volume Down**' button on the side of the phone until the Fastboot logo appears.

Back onto the Windows computer:

- Check if your phone is detected by writing the following in the command prompt:

```
fastboot devices
```

You should see your phone. If not, recheck the USB connectivity.

- Flash the patched image by using the command:

```
fastboot flash boot magisk_patched.img
```

- After flashing is successful, reboot the phone into the system:

```
fastboot reboot
```

This could take a few minutes.

- Once the phone is booted and set up, open the Magisk app.
- It should now say that Magisk is installed with whichever Magisk version you downloaded.
- Another way to verify is to download a '**Root Check**' app from the '**Play Store**'. If you do install the app, when the app prompts for root permissions, you will need to grant it.

3 Create Xiaomi Account

The problem for me when creating an account was the phone number.

Initially, I was not able to create an account using a phone number because I kept getting the error that the number was not valid even though it was, so I used the mail to create an account and added the phone number afterwards.

Then when I did create an account using mail, I was not able to find the place where I can add a phone number on the phone nor online. So I used the following link add my phone number:

<https://account.xiaomi.com/pass/serviceLogin?checkSafePhone=true>

which I got from the thread:

https://www.reddit.com/r/LineageOS/comments/1d6y974/how_do_i_add_phone_number_to_xiaomi_account/

Note that when I tried to do this initially on my laptop, it kept saying that the captcha I did to prove I am not a bot is incorrect, so I tried the link on my phone where it worked.

Steps to creating an account

1. On the phone go to 'Xiaomi Account' in settings.
2. Tap on create account and follow the on screen instructions. (Note that you will need a valid mail account for this).
3. If you do use a mail for creating an account, you will then need to link the phone number of your sim card to that Xiaomi account. To do this, go to the following link and link the phone number to your account: <https://account.xiaomi.com/pass/serviceLogin?checkSafePhone=true>. If it does not work, try the link on another device.

4 Reflashing the phone

Not done yet If you end up corrupting any of the partitions and is not able to boot into the system you can restore your device.

- Put phone into fastboot mode
- Use flash script